Clinical Risk Assessment Report: Patient 27

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 10.2% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.46x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Small Zone Emphasis (PET) (GLZLM_SZE.PET), which reduced the patient's absolute risk by 2.3%. Biologically, this feature reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

2. Key Pathological & Imaging Drivers (Elevating Risk)

No major risk-elevating features observed in the top drivers.

3. Protective / Mitigating Factors (Reducing Risk)

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: -0.3 [Avg (33th %ile)] (-2.3%)**

Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.17 [Avg (56th %ile)] (-0.9%)**

Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: -0.1 [Avg (61th %ile)] (-0.8%)**

Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

Squamous Cell Carcinoma Antigen (SCC_Ag) **Value: 0.8 [Avg (38th %ile)] (-0.7%)**

Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: -0.52 [Avg (31th %ile)] (-0.7%)**

Points to continuous, elongated bands of high-density or highly active tumor tissue.

Carcinoembryonic Antigen (CEA) **Value: 1.0 [Bottom 5%] (-0.6%)**

Elevated levels are strongly associated with increased tumor burden and potential for micrometastasis.

Tumor Local Contrast (PET) (GLCM_Contrast(=Variance).PET) **Value: -0.23 [Avg (60th %ile)] (-0.6%)**

Measures local variations and sharp transitions in density or metabolism, signifying an unstructured and variable tumor matrix.

Minimum Metabolic Activity (SUVmin) (CONVENTIONAL_SUVbwmin) **Value: -0.4 [Avg (42th %ile)] (-0.5%)**

Indicates the areas of lowest metabolic activity within the tumor, which may represent necrotic or poorly perfused regions.

Machine Learning Feature Attribution (SHAP Decision Path)